

Two Plastic-Reduction Quadrants and the Plastic–Elastic Decomposition of a Simple Shear

Technical note

14 July 2026

Abstract

For the simple shear $\mathbf{F} = \begin{pmatrix} 1 & 1.6 \\ 0 & 1 \end{pmatrix}$, the unit-shear breadth-first search reaches two points in the elastic domain. The shortest and longer routes produce visibly different matrices $\mathbf{M}$, and hence apparently different multiplicative decompositions $\mathbf{F} = \mathbf{F}_e \mathbf{F}_p$. The main observation of this note is that the two decompositions differ only by a 90° rotation of the intermediate basis. They have the same spatial elastic metric $\mathbf{B}_E$, the same reference plastic metric $\mathbf{C}_P$, the same elastic singular values, energy, and Cauchy stress. Thus the two elastic quadrants are physically equivalent unless an additional history or orientation variable fixes the intermediate frame. A direct comparison in the original coordinate frame does provide such a preference: the short-path plastic factor retains the positive horizontal shear of the total deformation, while the long-path factor looks like a vertical shear. The short factor is also much closer to the original $\mathbf{F}$. This is a meaningful frame-anchoring or continuity criterion, although the preference disappears if the intermediate frame is allowed to rotate by its admissible 90° lattice symmetry.

1 The two multiplicative decompositions

The accumulated integer matrix from plastic reduction acts on the right,

$$\mathbf{C}_E = \mathbf{M}^\top \mathbf{C} \mathbf{M}, \quad \mathbf{F}_e = \mathbf{F} \mathbf{M}. \quad (1)$$

Using the convention in the decomposition code,

$$\boxed{\mathbf{F} = \mathbf{F}_e \mathbf{F}_p, \quad \mathbf{F}_p = \mathbf{M}^{-1}.} \quad (2)$$

For the present $\mathbf{F}$, the shortest route has two unit shears, whereas the first route to the other elastic quadrant has three:

$$\text{short: } U_- U_-, \quad \mathbf{M}_s = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}, \quad (3)$$

$$\text{long: } U_- L_- U_+, \quad \mathbf{M}_\ell = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}. \quad (4)$$

Both factorizations reproduce the total deformation exactly. Their elastic and plastic factors are

$$\mathbf{F}_e^{(s)} = \begin{pmatrix} 1 & -0.4 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{F}_p^{(s)} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \quad (5)$$

$$\mathbf{F}_e^{(\ell)} = \begin{pmatrix} 0.4 & 1 \\ -1 & 0 \end{pmatrix}, \quad \mathbf{F}_p^{(\ell)} = \begin{pmatrix} 0 & -1 \\ 1 & 2 \end{pmatrix}. \quad (6)$$

Figure 1 puts these factors beside the matrices to which they should naturally be compared: $\mathbf{F}_p$ beside the original $\mathbf{F}$, and $\mathbf{F}_e$ beside the undeformed benchmark $\mathbf{I}$. To distinguish direct componentwise comparison from comparison after an admissible change of intermediate basis, define

$$\mathcal{G} = \{\mathbf{R}(k\pi/2) : k = 0, 1, 2, 3\} = \text{SO}(2) \cap \text{SL}(2, \mathbb{Z}). \quad (7)$$

The corresponding numerical tests are collected in Table 1.

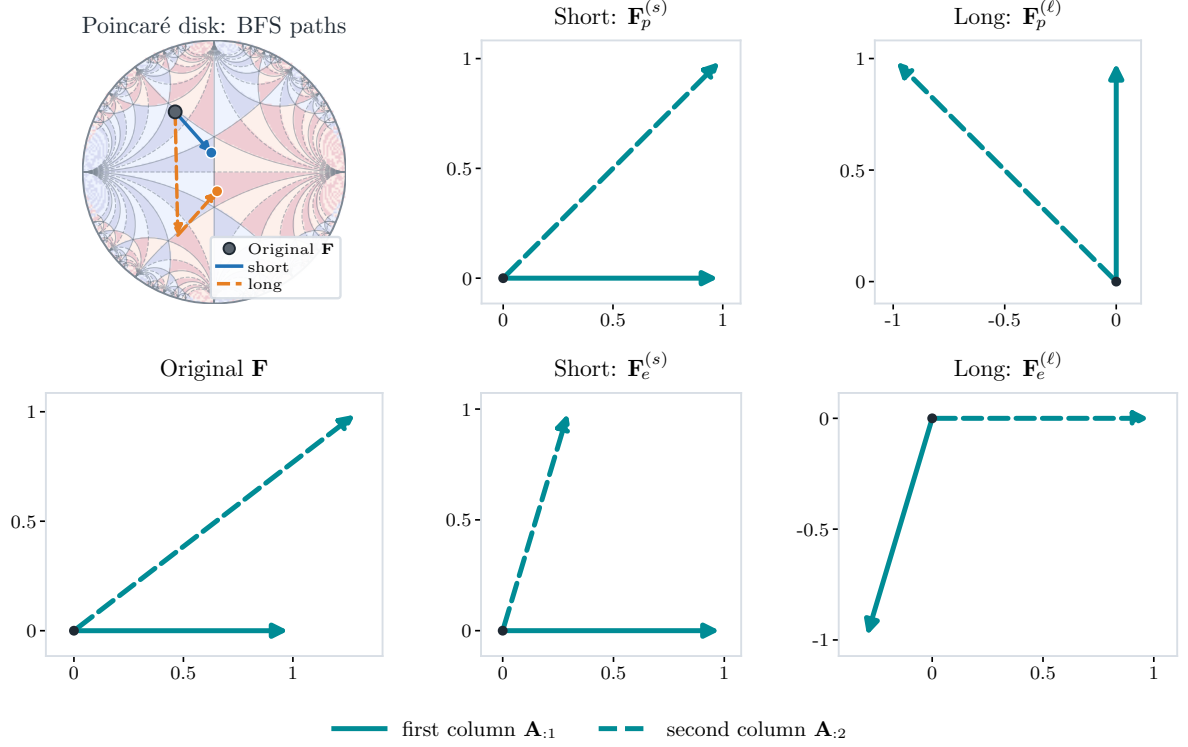


Figure 1: The original deformation and its two decompositions, together with the corresponding unit-shear BFS paths in the Poincaré disk. In every matrix panel the solid vector is the first column $\mathbf{A}_{:,1}$, and the dashed vector is the second column $\mathbf{A}_{:,2}$. The original $\mathbf{F}$ and the short-path $\mathbf{F}_p$ are both positive horizontal shears, whereas the long-path $\mathbf{F}_p$ appears vertically oriented. The differently oriented elastic factors compensate exactly, so both products $\mathbf{F}_e \mathbf{F}_p$ recover the same $\mathbf{F}$. Rotating the long-path intermediate basis back by 90° makes its plastic factor coincide with the short-path factor.

Table 1: Comparison with the original total deformation and the undeformed benchmark. “Fixed” means that all component matrices are read in the displayed axes; “aligned” allows the lattice rotations in $\mathcal{G}$.

Quantity	Original or benchmark	Shortest route	Longer route	Interpretation
Plastic matrix	$\mathbf{F} = \begin{pmatrix} 1 & 1.6 \\ 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 0 & -1 \\ 1 & 2 \end{pmatrix}$	Only the short factor retains the displayed upper-shear form
Fixed $\ \mathbf{F}_p - \mathbf{F}\ _F$	0	0.400	3.124	Strongly favors the short route
Aligned $\min_{\mathbf{R} \in \mathcal{G}} \ \mathbf{R}\mathbf{F}_p - \mathbf{F}\ _F$	0	0.400	0.400	Tie after intermediate-frame alignment
Reference-metric $\ \mathbf{C}_P - \mathbf{C}\ _F$	0	1.547	1.547	Objective comparison with the original $\mathbf{C}$ is tied
Elastic matrix	$\mathbf{I}$	$\begin{pmatrix} 1 & -0.4 \\ 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 0.4 & 1 \\ -1 & 0 \end{pmatrix}$	Only the short factor remains near the displayed axes
Fixed $\ \mathbf{F}_e - \mathbf{I}\ _F$	0	0.400	1.833	Favors the short route
Objective $\ \mathbf{B}_E - \mathbf{I}\ _F$	0	0.588	0.588	Spatial elastic comparison is tied
BFS depth	—	2	3	Favors the short route if each move has positive cost

1.1 Original-frame preference and its limit

With the intermediate axes identified with the original coordinate axes, the preference is unambiguous. The total deformation and the short-path plastic factor are positive upper-triangular shears of amplitudes 1.6 and 2, respectively. Their difference is only the residual shear 0.4. The long-path plastic factor has a different displayed orientation and is much farther from the total deformation. The elastic comparison tells the same story: the short-path factor is close to $\mathbf{I}$, whereas the long-path factor contains an additional quarter turn.

The limitation is that these direct matrix differences silently identify the intermediate vector space with the original reference and spatial coordinate spaces. The two integer matrices satisfy

$$\mathbf{M}_\ell = \mathbf{M}_s \mathbf{Q}, \quad \mathbf{Q} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \mathbf{R}(-\pi/2), \quad \mathbf{Q}^\top \mathbf{Q} = \mathbf{I}. \quad (8)$$

Consequently,

$$\mathbf{F}_e^{(\ell)} = \mathbf{F}_e^{(s)} \mathbf{Q}, \quad \mathbf{F}_p^{(\ell)} = \mathbf{Q}^\top \mathbf{F}_p^{(s)}. \quad (9)$$

The factors therefore change, but the rotation cancels from their product:

$$(\mathbf{F}_e^{(s)} \mathbf{Q})(\mathbf{Q}^\top \mathbf{F}_p^{(s)}) = \mathbf{F}_e^{(s)} \mathbf{F}_p^{(s)} = \mathbf{F}. \quad (10)$$

The apparently vertical long-path plastic shear is therefore exactly the short-path horizontal shear written in a rotated intermediate frame:

$$\mathbf{Q} \mathbf{F}_p^{(\ell)} = \mathbf{F}_p^{(s)}. \quad (11)$$

Accordingly, the fixed-frame plastic discrepancy selects the short path, but the discrepancy after lattice-compatible alignment does not:

$$\|\mathbf{F}_p^{(s)} - \mathbf{F}\|_F = 0.400, \quad \|\mathbf{F}_p^{(\ell)} - \mathbf{F}\|_F = 3.124, \quad (12)$$

$$\min_{\mathbf{R} \in \mathcal{G}} \|\mathbf{R} \mathbf{F}_p^{(s)} - \mathbf{F}\|_F = \min_{\mathbf{R} \in \mathcal{G}} \|\mathbf{R} \mathbf{F}_p^{(\ell)} - \mathbf{F}\|_F = 0.400. \quad (13)$$

The minimizing rotations are $\mathbf{I}$ for the short path and $\mathbf{Q}$ for the long path. The same conclusion follows from a comparison that never identifies the intermediate axes: since both routes have the same $\mathbf{C}_P$,

$$\|\mathbf{C}_P^{(s)} - \mathbf{C}\|_F = \|\mathbf{C}_P^{(\ell)} - \mathbf{C}\|_F = 1.547. \quad (14)$$

This is the standard rotational freedom of the intermediate configuration: for any $\mathbf{Q} \in \text{SO}(2)$, $\mathbf{F}_e \mathbf{F}_p = (\mathbf{F}_e \mathbf{Q})(\mathbf{Q}^\top \mathbf{F}_p)$. Here the admissible rotation is additionally lattice-compatible, $\mathbf{Q} \in \text{SO}(2) \cap \text{SL}(2, \mathbb{Z})$.

The quantities insensitive to this intermediate rotation are identical:

$$\mathbf{B}_E^{(s)} = \mathbf{F}_e^{(s)} \mathbf{F}_e^{(s)\top} = \mathbf{B}_E^{(\ell)} = \begin{pmatrix} 1.16 & -0.4 \\ -0.4 & 1 \end{pmatrix}, \quad (15)$$

$$\mathbf{C}_P^{(s)} = \mathbf{F}_p^{(s)\top} \mathbf{F}_p^{(s)} = \mathbf{C}_P^{(\ell)} = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}. \quad (16)$$

The two $\mathbf{F}_e$ have the same singular values, 1.219804 and 0.819804; the two $\mathbf{F}_p$ have singular values 2.414214 and 0.414214. They also give the same energy, 0.0864665, and the same Cauchy stress,

$$\boldsymbol{\sigma} = \begin{pmatrix} -0.0322271 & -0.0809712 \\ -0.0809712 & 0.0322271 \end{pmatrix}. \quad (17)$$

This explains why Cauchy stress cannot select between the quadrants: the spatial elastic metric $\mathbf{B}_E$, which carries the objective elastic stretch, is exactly the same.

2 What changes for even longer BFS paths?

At maximum BFS depth five, seven terminal paths are found. For this example, every accumulated matrix belongs to one of only four classes:

$$\mathbf{M} \in \{\mathbf{M}_s, -\mathbf{M}_s, \mathbf{M}_\ell, -\mathbf{M}_\ell\}. \quad (18)$$

The observed paths are summarized in Table 2.

Table 2: All terminal BFS paths through depth five.

Quadrant	Unit-shear word	Depth	Accumulated matrix class
1	$U_- U_-$	2	$\mathbf{M}_s$
1	$L_- U_+ U_+ L_-$	4	$-\mathbf{M}_s$
2	$U_- L_- U_+$	3	$\mathbf{M}_\ell$
2	$L_- U_+ L_+$	3	$\mathbf{M}_\ell$
2	$U_+ L_- U_+ L_+ L_+$	5	$\mathbf{M}_\ell$
2	$U_- L_+ U_- L_+ L_+$	5	$-\mathbf{M}_\ell$
2	$L_+ U_- L_+ L_+ L_+$	5	$-\mathbf{M}_\ell$

Thus the longer paths do not create new plastic or elastic stretch states in this case. Replacing $\mathbf{M}$ by $-\mathbf{M}$ gives $\mathbf{F}_e \mapsto -\mathbf{F}_e$ and $\mathbf{F}_p \mapsto -\mathbf{F}_p$. Their product, $\mathbf{C}_E$, $\mathbf{B}_E$, and $\mathbf{C}_P$ are unchanged. It is a 180° rotation of the oriented intermediate basis. Moving between the two elastic quadrants adds the 90° rotation in Equation (8).

This four-class conclusion is specific to the present generic metric and the paths searched here. A metric with additional lattice symmetries can have a larger stabilizer. The robust statement is that two factorizations are physically equivalent whenever their accumulated matrices differ by a consistently applied orthogonal change of intermediate basis.

If the intermediate frame is anchored to the original axes, the four matrix classes are no longer equally convenient. Write $d_P = \|\mathbf{F}_p - \mathbf{F}\|_F$ and $d_E = \|\mathbf{F}_e - \mathbf{I}\|_F$. Their scores are

$$\begin{array}{c|cccc} \mathbf{M} & \mathbf{M}_s & -\mathbf{M}_s & \mathbf{M}_\ell & -\mathbf{M}_\ell \\ \hline (d_P, d_E) & (0.400, 0.400) & (4.578, 2.857) & (3.124, 1.833) & (3.370, 2.227) \end{array} \quad (19)$$

Thus the anchored-frame criterion uniquely selects $\mathbf{M}_s$, including over the still longer paths. If rotations in $\mathcal{G}$ are first aligned away, all four classes again represent the same stretch state.

3 Basis-rotation context

“Basis transformation” is a reasonable name for the operation in the question. More precisely, it is a *passive orthogonal change of the reference basis*. If the new basis is represented by $\mathbf{Q}(\theta) = \mathbf{R}(\theta)$, the components transform as

$$\mathbf{F}(\theta) = \mathbf{F}(0)\mathbf{Q}(\theta), \quad \mathbf{C}(\theta) = \mathbf{Q}(\theta)^\top \mathbf{C}(0)\mathbf{Q}(\theta). \quad (20)$$

The deformation gradient has one reference-basis index and therefore contains one factor of $\mathbf{Q}$. The metric has two reference-basis indices and contains two factors. Its principal axes are directors rather than oriented vectors, so its component orbit closes after a basis rotation of π , while the component matrix $\mathbf{F}$ closes only after 2π .

For the example proposed in the question,

$$\mathbf{F}(0) = \begin{pmatrix} 1 & 0.4 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{C}(0) = \begin{pmatrix} 1 & 0.4 \\ 0.4 & 1.16 \end{pmatrix}, \quad (21)$$

one obtains

$$\mathbf{F}(\pi/2) = \begin{pmatrix} 0.4 & -1 \\ 1 & 0 \end{pmatrix}, \quad \mathbf{C}(\pi/2) = \begin{pmatrix} 1.16 & -0.4 \\ -0.4 & 1 \end{pmatrix}, \quad (22)$$

$$\mathbf{F}(\pi) = -\mathbf{F}(0), \quad \mathbf{C}(\pi) = \mathbf{C}(0). \quad (23)$$

Figure 2 shows the corresponding basis vectors and the closed orbit of $\mathbf{C}$ in the Poincare-disk coordinates used by the visualization code. At $\theta = \pi/2$, $\mathbf{C}$ has moved to the opposite elastic quadrant. At $\theta = \pi$, the oriented basis vectors have reversed, but the metric has returned to its initial value.

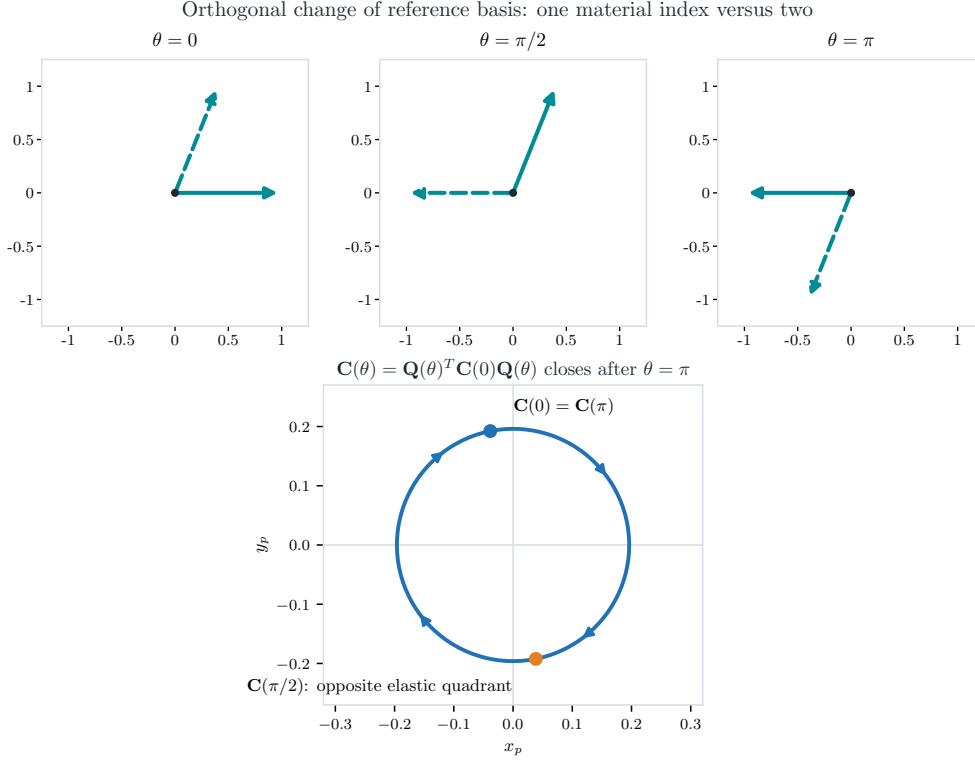


Figure 2: A passive rotation of the reference basis. The top row shows the columns of $\mathbf{F}(\theta)$. The bottom panel shows that $\mathbf{C}(\theta)$ completes a closed component-space orbit while the oriented columns of $\mathbf{F}$ have completed only a half turn. This is the same 90° intermediate-basis relation that connects the short and long plastic-reduction decompositions.

4 Interpretation and conclusion

Comparison with the original deformation gives two answers, depending on whether the intermediate frame is part of the state. If its axes are inherited from the original reference frame and are required to remain nearby, the short path is clearly preferable. Its plastic factor has the same positive horizontal-shear form as $\mathbf{F}$, differs from $\mathbf{F}$ by only 0.4, and leaves an elastic factor only 0.4 from $\mathbf{I}$. The long path inserts a quarter turn into the intermediate frame and is much less compatible in these fixed components. Equations (13) and (19) make this preference quantitative.

This suggests a useful selection rule: among admissible reductions, choose the one that minimizes $\|\mathbf{F}_e - \mathbf{I}\|_F$, or, equivalently for the present example, the one whose $\mathbf{F}_p$ remains closest to the total $\mathbf{F}$ in the anchored frame. In an incremental calculation, the more robust version is to choose the representative closest to the previously selected intermediate orientation. This suppresses unnecessary 90° or 180° jumps and makes the short path a natural continuity choice.

If the intermediate axes are not anchored, the horizontal-versus-vertical distinction is only a change of components. Applying the admissible rotation $\mathbf{Q}$ turns the long-path $\mathbf{F}_p$ exactly into the short-path $\mathbf{F}_p$. Their objective elastic and plastic metrics, singular values, energy, and Cauchy stress remain identical. Consequently the original $\mathbf{F}$ alone cannot create an objective preference between the two quadrants; it can select a preferred *representative* only after an orientation or continuity convention has been supplied.

A constitutive model can turn that convention into a physical preference by carrying plastic orientation or labeled slip systems as history variables. Alternatively, if every unit-shear event has the same positive dissipation, the BFS word length is itself a cost and independently favors the two-step short route over the three-step long route. Thus the short quadrant is defensible both because it preserves the anchored intermediate orientation and because it has lower path cost.